

🧠 AWARe: Mitigating Catastrophic Forgetting via Activation-Weighted Adaptive REtention

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Abstract

Multimodal Large Language Models (MLLMs) exhibit strong generalization and reasoning abilities due to large-scale multimodal pre-training. However, fine-tuning these models on downstream tasks often leads to catastrophic forgetting, where newly learned task-specific knowledge degrades previously acquired capabilities. This issue arises because gradient updates for new tasks overwrite parameters critical to prior knowledge, limiting the practical deployment of MLLMs. To address this challenge, we propose **Activation-Weighted Adaptive REtention (AWARe)**, a fine-tuning method that mitigates catastrophic forgetting by dynamically controlling parameter updates based on activation patterns. AWARe assigns activation-based importance scores to parameters, selectively freezing those essential for preserving prior capabilities while allowing less important parameters to adapt to new tasks. Importantly, AWARe operates *without modifying model architectures*, ensuring compatibility with existing inference engines. Extensive experiments demonstrate that AWARe effectively preserves upstream capabilities while achieving superior downstream performance compared to existing methods. Code is available at <https://github.com/kaln27/AWARe>.

1 Introduction

Multimodal Large Language Models (MLLMs) have emerged as powerful versatile agents, capable of solving complex tasks that require understanding and reasoning across both inputs of vision and language (Zhu et al., 2023; Dai et al., 2023; Bai et al., 2023; Liu et al., 2023; Lin et al., 2023; Chen et al., 2024b; Wang et al., 2024; Zhu et al., 2025a; Wang et al., 2025; Bai et al., 2025b,a). Typical MLLMs are composed of a pre-trained visual encoder, a large language model (LLM), and a connector module (Li et al., 2022; Dai et al., 2023; Liu et al., 2023). These models facilitate strong generalization abilities acquired from large-scale pre-training. Despite

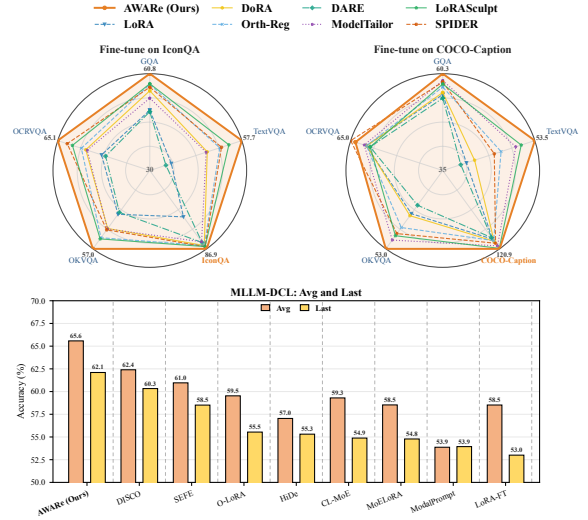


Figure 1: **Overall Performance of AWARe.** Top: radar plots showing comparison between our approach and various baselines under Single-Task Downstream Adaptation. Bottom: results on the MLLM-DCL continuous learning benchmark.

their impressive zero-shot capabilities, fine-tuning MLLMs on downstream tasks remains a standard practice to tailor models for specific domains or improve instruction-following performance (Dai et al., 2023; Bai et al., 2023; Liu et al., 2023). However, this adaptation process often comes at a cost: *catastrophic forgetting* (CF), where the model’s proficiency in previously learned upstream tasks degrades significantly as it optimizes for new objectives (McCloskey and Cohen, 1989; McClelland et al., 1995; Kirkpatrick et al., 2017; Li et al., 2024b). Nonetheless, this form of forgetting remains particularly severe in MLLMs due to the complex interplay between modalities and the high dimensionality of the parameter space (Sung et al., 2023; Luo et al., 2025, 2023; Zhai et al., 2024a; Shen et al., 2024; Jiang et al., 2024; Zhai et al., 2024b).

Existing approaches to mitigate catastrophic forgetting, such as experience replay (Riemer et al.,

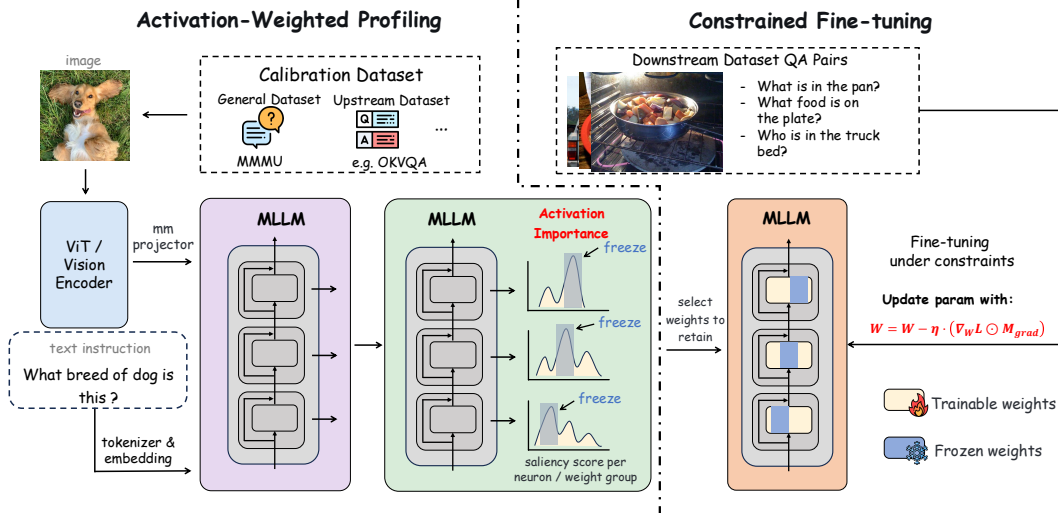


Figure 2: **Overview of AWARe.** Given a multimodal base model, AWARe identifies important neurons or weight groups using activation-based saliency on calibration samples, then freezes high-saliency weights and updates only low-saliency weights during fine-tuning.

2018; Chaudhry et al., 2019) or regularization-based methods (Kirkpatrick et al., 2017), often incur high computational overhead or struggle to scale to the billions of parameters in MLLMs. Parameter-Efficient Fine-Tuning (PEFT) methods like LoRA (Hu et al., 2022) reduce the trainable parameter count but do not inherently prevent the erosion of pre-trained knowledge, as recent studies show they can still disrupt reliable upstream features (Biderman et al., 2024; Zhu et al., 2024). Consequently, a critical challenge arises: *How can we enable MLLMs to adapt effectively to downstream tasks while robustly preserving their pre-trained capabilities without prohibitive costs?*

Our work is primarily inspired by the principle of activation-aware parameter saliency, most notably demonstrated by Activation-aware Weight Quantization (AWQ) (Lin et al., 2024). AWQ shows that a small fraction of weights associated with large activation magnitudes are disproportionately important for preserving model behavior under quantization. Wanda reaches a complementary conclusion in pruning: static weight magnitude alone is insufficient for LLMs, while combining weights with input activation statistics gives a simple and effective estimate of functional importance (Sun et al., 2024). Recent model merging studies further suggest that activation-guided consensus can reduce parameter interference when combining specialized models (Nobari et al., 2025; Yao et al., 2025). Together, these findings indicate that activations provide a dynamic, data-dependent view of which internal pathways support stable behavior, beyond what

weight-centric criteria such as magnitude pruning can capture (Han et al., 2015; Frankle and Carbin, 2019). We therefore repurpose activation saliency from compression and merging to forgetting mitigation.

Building on this motivation, we introduce **AWARe**, short for **Activation-Weighted Adaptive REtention**, a novel framework designed to surgically mitigate catastrophic forgetting. AWARe has two phases: *Knowledge Profiling*, which estimates neuron saliency from a small calibration set, and *Constrained Fine-tuning*, which freezes the most salient parameters during downstream training. When upstream data are unavailable, we can use a general-purpose calibration set such as MMMU (Yue et al., 2024); as shown by the AWARe (MMMU) results in Table 1, this remains highly effective while adding only a negligible profiling cost.

We validate our approach through comprehensive experiments on IconQA (Lu et al., 2021), COCO-Caption (Lin et al., 2014), and MLLM-DCL (Zhao et al., 2025) (Continuous learning) benchmarks. AWARe achieves strong stability-plasticity trade-offs, and freezing the top 30% of self-attention parameters is often sufficient to preserve upstream knowledge while keeping downstream performance competitive.

Our main contributions are summarized as follows:

- 1 **AWARe Framework.** We introduce Activation-Weighted Adaptive REtention (AWARe), a simple activation-based method for preserving crit-

ical upstream knowledge during downstream adaptation in MLLMs.

- ② **Efficiency and Simplicity.** Our approach requires no architectural changes and updates only a small subset of parameters (e.g., $\sim 17.5\%$; see Section C), avoiding replay buffers and extra modules.
- ③ **Comprehensive Validation.** We show through ablations and benchmarks that activation-based saliency is effective and that AWARE preserves generalization across diverse settings.

2 Related Work

2.1 Catastrophic Forgetting and Continual Learning

Fine-tuning serves as a pivotal mechanism for adapting Multimodal Large Language Models (MLLMs) to downstream tasks. However, deep learning models often suffer from catastrophic forgetting (McCloskey and Cohen, 1989; McClelland et al., 1995), where previously learned knowledge is lost when acquiring new skills. Various continual learning algorithms have been proposed to address this issue, generally encompassing rehearsal-based, regularization-based (Kirkpatrick et al., 2017; Lopez-Paz and Ranzato, 2017), and architecture-based approaches (Houlsby et al., 2019; Lester et al., 2021). While established, many traditional methods rely on full-model fine-tuning or second-order statistics, making them computationally prohibitive for large-scale models.

2.2 Parameter-Efficient Fine-Tuning for Mitigation

In the era of large foundation models, the challenge shifts towards maintaining the model’s strong generalization and "Open-World Stabilization" after downstream adaptation (Ni et al., 2023; Zhai et al., 2024a). Recent efforts have explored parameter-efficient solutions to mitigate forgetting. Reparameterization methods such as DoRA (Liu et al., 2024) and LoRA-based extensions like LoRAMoE (Dou et al., 2024) aim to enhance learning capacity while addressing generalization loss, whereas LoRASculpt (Liang et al., 2025) prunes redundant parameters via sparse updates. Regularization techniques have also been adapted; for instance, Orth-Reg (Hu et al., 2024) encourages fine-tuned features to remain orthogonal to pre-trained representations. CorDA (Yang et al., 2024) builds task-aware

adapters through context-guided weight decomposition to preserve world knowledge. Selective tuning approaches, such as SPIDER (Huang et al., 2025) and Model Tailor (Zhu et al., 2024), leverage gradient or saliency analysis to construct sparse updates, aiming to protect critical pre-trained features. However, these criteria often rely on static information or gradient magnitudes, overlooking the dynamic nature of neuronal activations during forward propagation.

Recognizing these limitations, we propose AWARE. Unlike additive methods that introduce extra parameters or selective tuning that relies on static priors, AWARE utilizes task-induced activation distributions to measure parameter importance. By adaptively constraining high-activation neurons based on upstream statistics, AWARE effectively balances the preservation of core capabilities with the flexibility needed for downstream task adaptation.

3 Methodology

3.1 Problem Formulation

A Multimodal Large Language Model (MLLM) $\mathcal{M}_\theta$ generally comprises three core components: a vision encoder $\mathcal{V}$ (e.g., CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023)), a large language model $\mathcal{L}$ (e.g., LLaMA (Touvron et al., 2023b,a) or Vicuna (Chiang et al., 2023)), and a connector module ρ that bridges the visual and textual modalities. The standard paradigm involves aligning the pre-trained vision encoder representations with the LLM’s embedding space via the connector, enabling the model to process multimodal inputs effectively.

Consider a Multimodal Large Language Model (MLLM) $\mathcal{M}_\theta$ parameterized by θ . We assume the model has been initially trained on a foundational set of upstream tasks $\mathcal{T}_{base}$. In a continual instruction tuning setting, the model is sequentially exposed to a stream of downstream tasks, denoted as $\mathcal{T}_{seq} = \{\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_N\}$. Our objective is to incrementally learn each new task $\mathcal{T}_t$ at step t by updating θ , while mitigating the *catastrophic forgetting* (McCloskey and Cohen, 1989; McClelland et al., 1995) of both the foundational capabilities from $\mathcal{T}_{base}$ and the previously acquired knowledge from tasks $\{\mathcal{T}_1, \dots, \mathcal{T}_{t-1}\}$. As illustrated in Figure 3, we achieve this by partitioning the parameters into a fixed set θ_{fixed} , which captures and preserves core established features, and a plastic set θ_{active} used

for acquiring new task-specific knowledge.

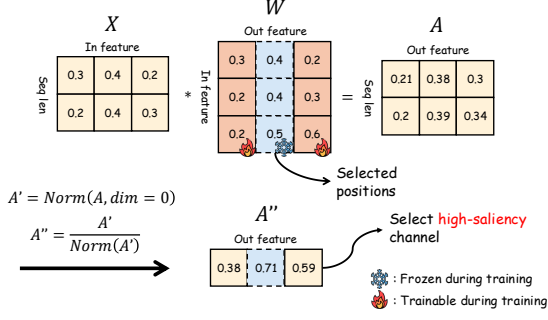


Figure 3: Overview of AWARe. For each target linear layer, we profile the activation matrix produced by calibration samples, normalize the activation statistics across sequence and feature dimensions, and obtain a saliency score for each output channel. High-saliency channels are selected and frozen to preserve pretrained knowledge, while the remaining channels stay trainable for downstream adaptation.

3.2 Activation-Based Saliency Estimation

To identify important parameters for retention, we utilize a calibration set $\mathcal{S}_{cal}$ composed of representative samples from each upstream task’s train datasets ($\mathcal{T}_{up}$). Specifically, we randomly sample several instances from each upstream task. In scenarios where access to upstream data is restricted, we demonstrate that utilizing a comprehensive general benchmark, such as MMMU (Yue et al., 2024), as a proxy for $\mathcal{S}_{cal}$ is equally effective. We empirically demonstrate the robustness of model performance to the calibration set size in our ablation studies at Section 4.6. Crucially, the profiling incurs negligible overhead, requiring only a single forward pass on a small sample set. For a target linear layer with weight matrix $\mathbf{W} \in \mathbb{R}^{d_{out} \times d_{in}}$, we analyze its activation tensor $\mathbf{A} \in \mathbb{R}^{B \times L \times d_{out}}$, where $i \in \{1, \dots, B\}$, $j \in \{1, \dots, L\}$, and $k \in \{1, \dots, d_{out}\}$ index the batch size, context length and hidden dimension, respectively.

The saliency estimation follows a three-step aggregation and normalization process. First, we compute the L_2 -norm of activations along the sequence length dimension for each sample j and neuron k :

$$a'_{i,k} = \sqrt{\sum_{j=1}^L a_{i,j,k}^2} \quad (1)$$

Second, to ensure comparability across samples and prevent those with outlier-scale activations from biasing the estimations, we apply per-sample

L_2 -normalization across the hidden dimension:

$$a''_{i,k} = \frac{a'_{i,k}}{\|\mathbf{a}'_i\|_2} = \frac{a'_{i,k}}{\sqrt{\sum_{k'=1}^{d_{out}} a'_{i,k'}^2}} \quad (2)$$

where $\mathbf{a}'_i = [a'_{i,1}, \dots, a'_{i,d_{out}}]^\top$ is the vector of sequence-aggregated activations for sample i . This normalization ensures that the saliency is determined by the *relative* importance of neurons within each context rather than absolute activation scale, which is crucial for identifying neurons that consistently capture structural features across diverse upstream tasks. This refinement distinguishes our approach from simple magnitude-based methods and explains the superior performance observed in Table 3. Finally, the saliency score for each neuron k is obtained by averaging across the batch:

$$s_k = \frac{1}{B} \sum_{i=1}^B a''_{i,k} \quad (3)$$

The resulting saliency vector $\mathbf{s} \in \mathbb{R}^{d_{out}}$ reflects the relative importance of each output neuron in preserving upstream task capabilities.

3.3 Adaptive Parameter Retention

Guided by the saliency scores, we introduce a retention ratio $\rho \in (0, 1)$ as a hyperparameter to control the fraction of parameters to be frozen. We rank output neurons across all target linear layers by their saliency scores and identify the global index set $\mathcal{I}$ containing the top- ρ fraction of neurons (e.g., top 30% or 10%).

Since these salient neurons characterize the core upstream knowledge, we freeze their corresponding parameters in the weight matrix. Specifically, we construct a binary gradient mask $\mathbf{M}_{grad} \in \{0, 1\}^{d_{out} \times d_{in}}$ such that:

$$\begin{aligned} (M_{grad})_{k,:} &= \mathbf{0} \quad \text{if } k \in \mathcal{I}, \\ (M_{grad})_{k,:} &= \mathbf{1} \quad \text{otherwise} \end{aligned} \quad (4)$$

$$M_{ret} = \mathbf{1} - M_{grad}$$

where $\mathbf{0}$ and $\mathbf{1}$ are row vectors of size d_{in} . During downstream optimization, the parameters associated with salient regions are held constant via masked gradient updates:

$$\mathbf{W}^{(t+1)} = \mathbf{W}^{(t)} - \eta \cdot (\nabla_{\mathbf{W}} \mathcal{L}_{down} \odot \mathbf{M}_{grad}) \quad (5)$$

where η is the learning rate and $\odot$ denotes the Hadamard product. In practice, the PyTorch implementation achieves this by utilizing a training-time wrapper, `AwareLinear`, which decomposes

the linear layer into active and frozen components to efficiently manage gradient updates without altering the underlying operator’s logic. This wrapper is removed post-training, leaving the original model configuration intact. This row-wise freezing mechanism ensures that the most influential output features of the pre-trained LLaVA model remain intact while allowing less critical regions to adapt to new tasks. As shown in Figure 4, the distribution of these salient neurons varies across transformer layers and projection types.

3.4 Selection of Target Linear

We specifically apply AWARE to the linear projection layers within the self-attention mechanism (i.e., q_proj, k_proj, v_proj) and the linear layers within the multimodal projector (mm_projector). This selection is informed by recent research (Zhu et al., 2025b) which demonstrates that updating self-attention projections tends to cause significantly less catastrophic forgetting of pre-existing knowledge compared to the multilayer perceptron (MLP) blocks. By focusing our retention strategy on those linear, we balance the preservation of core cross-modal reasoning capabilities with the flexibility needed for downstream task adaptation. An ablation analysis is provided in Section 4.5. Consistent with this finding, all other components of the LLM, including MLP blocks and the output projection ($\mathbf{W}_o$), are kept entirely frozen during adaptation. This configuration not only maximizes parameter efficiency but also minimizes the risk of distorting the pre-trained feature space in regions less critical for downstream alignment. The complete procedure for AWARE is summarized in Algorithm 1.

4 Experiments

In this section, we evaluate the effectiveness of our proposed method by comparing it with several state-of-the-art baselines on multimodal downstream tasks. We also conduct extensive ablation studies to analyze the impact of key components and hyperparameters.

4.1 Experimental Setup

Datasets and Tasks. We use LLaVA-v1.5-7b (Liu et al., 2023) as the base model for training. The model has been pre-trained on several upstream tasks, including OKVQA (Marino et al., 2019), OCRVQA (Mishra et al., 2019), GQA (Hudson and Manning, 2019), and TextVQA (Singh et al.,

Algorithm 1 AWARE

Require: Pre-trained model $\mathcal{M}_\theta$, Upstream task $\mathcal{T}_{up}$, Downstream task $\mathcal{T}_{down}$, Retention ratio ρ

Ensure: Fine-tuned model $\mathcal{M}_\theta$

- 1: **Phase 1: Knowledge Profiling**
 - 2: Sample calibration set $\mathcal{S}_{cal} \sim \mathcal{T}_{up}$
 - 3: **for** each target layer l in $\mathcal{M}_\theta$ **do**
 - 4: Perform forward pass on $\mathcal{S}_{cal}$ to obtain activations $\mathbf{H}^{(l)}$
 - 5: Calculate neuron saliency $\mathbf{s}^{(l)}$ using Equations (1) to (3)
 - 6: **end for**
 - 7: $\mathcal{I} \leftarrow \text{GlobalTopIndices}(\{\mathbf{s}^{(l)}\}_l, \rho)$
 - 8: **for** each target layer l in $\mathcal{M}_\theta$ **do**
 - 9: Extract layer-specific retained indices $\mathcal{I}^{(l)}$ from $\mathcal{I}$
 - 10: Generate row-wise gradient mask $\mathbf{M}_{grad}^{(l)}$ via Equation (4)
 - 11: **end for**
 - 12: **Phase 2: Constrained Fine-tuning**
 - 13: **while** not converged on $\mathcal{T}_{down}$ **do**
 - 14: $\mathcal{L} \leftarrow \text{ComputeLoss}(\mathcal{M}_\theta, \mathcal{T}_{down})$
 - 15: Update parameters: $\theta \leftarrow \theta - \eta \cdot (\nabla_\theta \mathcal{L} \odot \mathbf{M}_{grad})$
 - 16: **end while**
-

2019). For downstream evaluation, we use IconQA (Lu et al., 2021) for VQA and COCO-Caption (Lin et al., 2014) for image captioning, where captioning performance is measured by CIDEr. For continual instruction tuning, we evaluate LLaVA-v1.5-7b on MLLM-DCL (Zhao et al., 2025), a five-task sequence covering remote sensing (RS), medical VQA (Med), autonomous driving (AD), science reasoning (Sci), and financial understanding (Fin). We provide additional experiments on Qwen2.5-VL-7B (Bai et al., 2025b) in Section B.

Implementation Details. During training, we freeze the vision tower to preserve the pre-trained visual representations. The complete training configuration and hyperparameter settings are provided in Section D.

4.2 Baselines

We evaluate AWARE under two complementary settings. First, in single-task downstream adaptation, the model learns one new downstream task and is evaluated for both target-task learning and forgetting on upstream tasks. In this setting, we compare against established parameter-efficient

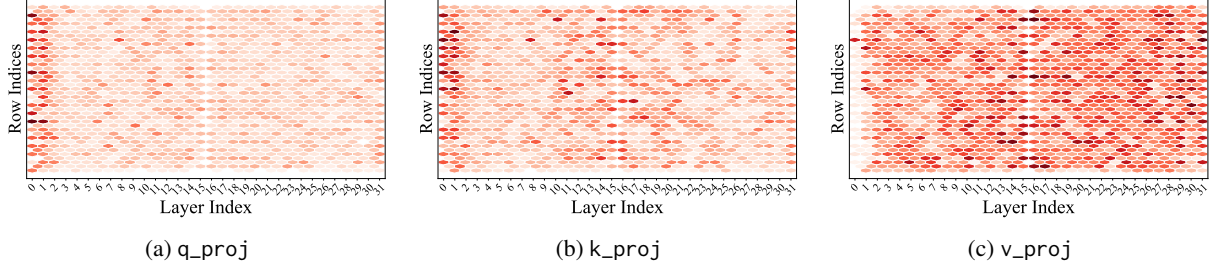


Figure 4: Layer-wise distribution of high-activation neurons across different projection types. The heatmaps visualize indices identified for freezing (top 30% global activation). Notably, q_proj and k_proj show high concentration in early layers, whereas v_proj exhibits a more uniform distribution, suggesting value information is distributed more evenly across the depth of the network.

Methods	IconQA								COCO-Caption							
	OKVQA	OCRQA	GQA	TextVQA	Target	$\mathcal{R}$	$\mathcal{E}$	$\mathcal{H}$	OKVQA	OCRQA	GQA	TextVQA	Target	$\mathcal{R}$	$\mathcal{E}$	$\mathcal{H}$
Zero-shot	57.99	66.20	61.93	58.23	23.18	100.0	28.9	44.9	57.99	66.20	61.93	58.23	40.40	100.0	42.3	59.4
Full-FT †	0.04	0.00	0.00	0.46	80.15	0.2	100.0	0.4	0.00	0.00	0.00	0.04	95.59	0.0	100.0	0.0
LoRA	45.06	48.40	49.22	36.47	63.65	73.3	79.4	76.2	44.92	59.90	54.88	39.75	110.27	81.6	115.4	95.6
DoRA	49.94	54.40	55.39	47.26	84.48	84.7	105.4	93.9	45.38	59.75	55.40	41.38	112.60	82.6	117.8	97.1
Orth-Reg	53.12	56.10	57.43	51.00	84.52	89.1	105.5	96.6	48.11	59.85	56.83	46.69	111.87	86.5	117.0	99.5
DARE	44.39	46.85	48.75	34.84	82.28	71.5	102.7	84.3	42.98	58.65	53.92	38.61	108.57	79.5	113.6	93.5
Model Tailor	50.07	53.85	53.04	46.92	81.76	83.4	102.0	91.8	50.99	60.60	58.54	49.65	117.64	89.9	123.1	103.9
LoRASculpt	<u>53.52</u>	59.50	<u>57.63</u>	<u>53.76</u>	85.26	<u>91.8</u>	106.4	<u>98.6</u>	49.99	58.65	57.63	<u>50.73</u>	<u>120.35</u>	88.8	<u>125.9</u>	<u>104.2</u>
SPIDER	50.41	61.49	56.54	51.54	85.29	90.0	106.4	97.5	49.50	65.00	58.14	45.33	114.74	89.2	120.0	102.3
AWARE	56.95	65.10	60.83	57.66	86.92	98.4	108.4	103.2 $\uparrow 4.6$	53.01	63.45	60.32	53.46	120.94	94.2	126.5	108.0 $\uparrow 3.8$
AWARE (MMU)	56.86	63.35	60.93	56.77	85.23	97.4	106.3	101.7 $\uparrow 3.1$	53.25	63.12	60.38	52.74	121.22	93.9	126.8	107.9 $\uparrow 3.7$

Table 1: **Comparison with State-of-the-Art Fine-Tuning Solutions on LLaVA-v1.5-7B** for IconQA and COCO-Caption. The optimal and sub-optimal results are denoted by boldface and underlining. $\uparrow$ means improved accuracy compared with the sub-optimal results. †Specifically, the near-zero performance of Full-FT on upstream tasks reflects extreme catastrophic forgetting due to overfitting to the target task. More details about the metrics are provided in Section 4.3. Training setting are detailed in Section D.

fine-tuning and anti-forgetting baselines, including **Full-FT** (Hu et al., 2022), **LoRA** (Hu et al., 2022), **DoRA** (Liu et al., 2024), **DARE** (Yu et al., 2024), **Orth-Reg** (Hu et al., 2024), **Model Tailor** (Zhu et al., 2024), **LoRASculpt** (Liang et al., 2025), and **SPIDER** (Huang et al., 2025). Second, in continual instruction tuning, the model learns a sequence of sub-tasks and is evaluated after each training stage. For this setting, we compare against continual learning baselines including **ModalPrompt** (Zeng et al., 2025), **SEFE** (Chen et al., 2025), **HiDe-LLaVA** (Guo et al., 2025a), **CL-MoE** (Huai et al., 2025), **O-LoRA** (Wang et al., 2023), **MoELoRA** (Chen et al., 2024a), and **DISCO** (Guo et al., 2025b). Detailed descriptions for all baselines are provided in Section E.

4.3 Single-Task Downstream Adaptation

We evaluate the performance of all methods by decomposing continual learning capabilities into two core components: **Knowledge Retention** ($\mathcal{R}$) and **Learning Efficiency** ($\mathcal{E}$). To provide a balanced assessment, we report their Harmonic Mean ($\mathcal{H}$):

$$\mathcal{H} = \frac{2 \cdot \mathcal{R} \cdot \mathcal{E}}{\mathcal{R} + \mathcal{E}} \quad (6)$$

The components are defined as follows:

- $\mathcal{R}$: A stability metric measuring the preservation of performance on upstream tasks after fine-tuning. It is calculated as the average ratio of the fine-tuned accuracy to the base (zero-shot) accuracy:

$$\mathcal{R} = \frac{1}{|\mathcal{T}_{up}|} \sum_{i \in \mathcal{T}_{up}} \frac{Acc_{i,final}}{Acc_{i,base}} \quad (7)$$

where $Acc_{i,base}$ is the model’s initial performance before fine-tuning. A sharp drop in $\mathcal{R}$ indicates catastrophic forgetting.

- $\mathcal{E}$: A plasticity metric measuring the model’s adaptation to the new downstream task. To normalize against model capacity, it is defined relative to the Full Fine-Tuning (Full-FT) baseline:

$$\mathcal{E} = \frac{Acc_{down,Method}}{Acc_{down,Full-FT}} \quad (8)$$

where $Acc_{down,Full-FT}$ represents the performance upper bound achievable by Full-FT.

As shown in Table 1, our method achieves a superior harmonic balance $\mathcal{H}$ between plasticity and stability. Compared to other methods, our approach maintains high retention rates $\mathcal{R}$ while achieving efficiency $\mathcal{E}$. Notably, when upstream task data is

Method	Avg						Last					
	RS $\uparrow$	Med $\uparrow$	AD $\uparrow$	Sci $\uparrow$	Fin $\uparrow$	Average $\uparrow$	RS $\uparrow$	Med $\uparrow$	AD $\uparrow$	Sci $\uparrow$	Fin $\uparrow$	Average $\uparrow$
ModalPrompt	53.19	45.73	40.78	41.82	87.82	53.87	53.63	45.68	40.77	41.81	87.82	53.94
HiDe	75.41	48.71	37.06	42.49	81.55	57.04	74.31	48.95	33.21	38.54	81.55	55.31
LoRA-FT	73.34	47.97	38.68	45.16	87.45	58.52	69.65	41.59	25.43	40.88	87.45	53.00
MoELoRA	77.19	47.02	37.30	44.40	86.75	58.53	77.54	41.85	27.62	40.13	86.75	54.78
CL-MoE	73.51	49.97	38.55	45.73	88.74	59.30	71.34	46.84	26.33	41.17	88.74	54.88
O-LoRA	76.12	47.38	40.97	46.02	87.15	59.53	74.64	44.42	30.02	41.47	87.15	55.54
SEFE	77.38	51.41	44.24	44.95	86.82	60.96	77.26	50.37	37.21	40.87	86.82	58.51
DISCO	77.41	48.04	48.70	48.65	89.22	62.40	76.49	44.48	44.84	46.61	89.22	60.33
AWARE	79.88	55.42	51.38	49.79	91.45	65.58	78.96	47.01	46.94	46.13	91.45	62.10

Table 2: **Detailed continuous instruction tuning performance on the MLLM-DCL benchmark** (Zhao et al., 2025). *Avg* reports the average score for each subtask across all later evaluation points after that subtask is learned, while *Last* reports the final score on each subtask after training on the last subtask.

restricted, we used the MMMU general dataset as a calibration dataset and still achieved relatively good results, outperforming both LoRASculpt and SPIDER. We further verify that AWARE is not limited to the LLaVA-v1.5 backbone by conducting an additional Qwen2.5-VL base-model experiment; these results are reported in Section B.

4.4 Continual Instruction Tuning on MLLM-DCL

To further demonstrate the effectiveness of our proposed method in sequential learning scenarios, we evaluate AWARE on the MLLM-DCL benchmark (Zhao et al., 2025). This benchmark is specifically designed to assess continual instruction tuning for multimodal large language models. As shown in Table 2, AWARE achieves the highest average per-task performance (*Avg*) and the best final performance after the last task (*Last*), improving the overall averages by 3.18 and 1.77 points over the strongest baseline, respectively. The detailed CL results are provided in Section A. Overall, AWARE effectively balances plasticity and stability, confirming its superiority in continual instruction tuning settings.

4.5 Ablation Studies

We conduct ablation studies to investigate the contribution of different components and hyperparameter choices in our framework.

Effectiveness of Activation-based Selection.

We compare our activation-based parameter selection strategy against weight norm-based selection, a hybrid strategy ($0.3 \times \text{Weight} + 0.7 \times \text{Activation}$), and a baseline using random parameter selection at the same ratio (30%). As shown in Table 3, random selection significantly fails to protect the upstream knowledge, with the average performance on upstream tasks (*Source Avg*) dropping to 49.94 on

the IconQA task. In contrast, our activation-based selection consistently achieves the best balance, outperforming both random and weight-based selection in both *Source Avg* and *Target* scores.

Strategy	IconQA		COCO-Caption	
	Source Avg	Target	Source Avg	Target
Random Selection	49.94	86.32	49.46	116.31
Weight Norm	58.51	85.51	53.52	115.19
$0.3W + 0.7A$	58.69	84.66	53.90	115.24
Activation	60.13	86.92	57.56	120.94

Table 3: **Ablation Study on Parameter Selection Strategy.** We compare different methods for selecting parameters to retain: random selection (Random Selection), weight magnitude (Weight Norm), activation magnitude (Activation), and a hybrid approach ($0.3W + 0.7A$).

Impact of Selection Ratio and Strategy. We explore the effect of varying the selection ratio (from 1% to 90%) and comparing Layer-Balanced versus Global-Highest selection strategies. As shown in Table 4, the Global-Highest selection strategy at a 30% ratio achieves the optimal balance between stability and plasticity, yielding the highest harmonic mean ($\mathcal{H} = 103.2$) and superior knowledge retention ($\mathcal{R} = 98.4$).

Ratio	OKVQA	OCRVQA	GQA	TextVQA	Target	$\mathcal{R}$	$\mathcal{E}$	$\mathcal{H}$
Layer balanced selection								
1%	55.33	64.00	60.17	54.39	82.44	95.7	102.9	99.2
10%	55.33	65.20	59.64	55.16	86.84	96.3	108.3	102.0
30%	56.28	62.75	60.62	56.13	85.29	96.5	106.4	101.2
40%	56.47	63.85	60.46	56.94	86.13	97.3	107.5	102.1
50%	57.33	63.95	60.80	57.19	86.21	97.9	107.6	102.5
70%	57.69	64.20	61.77	57.54	85.67	98.7	106.9	102.6
90%	57.65	63.70	62.01	57.92	84.07	98.7	104.9	101.7
Global highest selection								
1%	55.26	63.40	59.89	54.96	82.65	95.6	103.1	99.2
10%	55.88	63.25	59.59	55.39	86.13	95.8	107.5	101.3
30%	56.95	65.10	60.83	57.66	86.92	98.4	108.4	103.2
40%	57.66	64.35	61.47	57.77	86.64	98.7	108.1	103.2
50%	57.71	64.75	61.87	57.86	85.97	99.1	107.3	103.0
70%	57.78	64.35	62.03	57.84	80.64	99.0	100.6	99.8
90%	58.13	65.55	61.89	58.33	54.62	99.8	68.1	81.0

Table 4: **Ablation Study on Selection Ratio.** We evaluate the performance of our method with different ratios of parameters selected for retention on the IconQA task.

Calibration Dataset Composition. We exam-

ine how the composition of the calibration dataset influences the model’s resistance to catastrophic forgetting on the IconQA task. As shown in Table 5, using any individual upstream task (e.g., OKVQA, OCRVQA, GQA, or TextVQA) as the calibration source consistently preserves knowledge, with relatively small performance variations across different configurations. However, our comprehensive calibration approach (ALL), which integrates samples from all upstream sources, yields the most balanced and superior overall performance ($\mathcal{H} = 103.2$). This demonstrates that while AWARe is robust to the specific choice of calibration task, a diverse mixture of upstream data ensures the most effective protection across the model’s entire functional landscape.

Calibration Task	OKVQA	OCRVQA	GQA	TextVQA	Target	$\mathcal{R}$	$\mathcal{E}$	$\mathcal{H}$
OKVQA	55.17	63.96	58.56	54.72	86.97	95.1	108.5	101.4
OCRVQA	54.76	65.16	58.28	54.45	85.85	95.2	107.1	100.8
GQA	55.92	64.22	60.12	54.97	86.21	96.3	107.6	101.6
TextVQA	55.56	64.40	59.78	56.11	85.78	96.5	107.0	101.5
ALL	56.95	65.10	60.83	57.66	86.92	98.4	108.4	103.2

Table 5: **Ablation Study on Calibration Dataset Composition.** We investigate how using different subsets of upstream data for activation statistics calculation affects performance on the IconQA task.

Training Targets. We investigate the impact of applying our method to different components within the model architecture. As illustrated in Figure 5, targeting both the self-attention projections and the multimodal (mm_projector) yields the most favorable balance between plasticity and stability. Specifically, the mm_projector is essential for learning new downstream concepts; Locking it limits adaptation and causes a roughly 10-point decline in target performance. Furthermore, we observe that fine-tuning MLP blocks results in a severe performance collapse on upstream tasks (dropping below 30-point), even with our adaptive protection. This reinforces the hypothesis that MLP weights house the bulk of the model’s fundamental general knowledge, whereas self-attention projections provide a more flexible landscape for specialized task adaptation.

4.6 Sensitivity to Calibration Datasets

To assess the statistical significance and robustness of our results, we evaluate the sensitivity of AWARe to the randomness of the calibration set. We repeat the Knowledge Profiling phase over three independent runs with different random seeds for sample selection. As shown in Table 6, the performance metrics exhibit minimal standard deviation across runs, demonstrating that the identi-

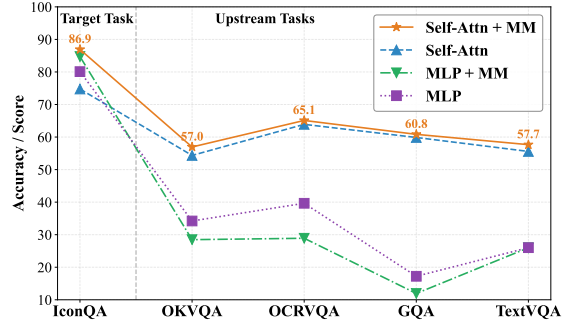


Figure 5: **Ablation Study on Training Target.** Performance comparison across different architecture components on the IconQA and upstream tasks. Applying AWARe to both self-attn and mm_projector achieves the best stable-plastic trade-off.

fied salient neurons are consistent and statistically stable artifacts of the model structure, rather than noise from specific data samples. Moreover, increasing the calibration size from 200 to 400 samples yields negligible performance differences, confirming that 200 samples are statistically sufficient to robustly estimate parameter importance.

Calibration Size	OKVQA	OCRVQA	GQA	TextVQA	Target	$\mathcal{R}$	$\mathcal{E}$	$\mathcal{H}$
200	56.95	65.10	60.83	57.66	86.92	98.4	108.4	103.2
200	56.88	64.45	61.20	58.04	86.57	98.5	108.0	103.0
200	56.49	64.50	60.49	58.15	86.58	98.1	108.0	102.8
400	56.37	64.38	60.58	57.77	86.20	97.9	107.5	102.5
400	56.74	64.40	60.38	57.17	86.69	97.7	108.2	102.7
400	57.04	64.62	60.47	56.77	86.35	97.8	107.7	102.5
Std. Dev.	0.24	0.25	0.28	0.48	0.23	0.30	0.30	0.30

Table 6: **Sensitivity Analysis on Calibration Datasets.** We evaluate the performance of AWARe on IconQA task across different calibration set sizes (200 and 400) and random sample 3 times.

5 Conclusion

In this paper, we introduce AWARe, an activation-weighted framework for mitigating catastrophic forgetting in Multimodal Large Language Models. AWARe first profiles task-induced activations to identify salient neurons, then freezes these regions during downstream fine-tuning to preserve pre-trained capabilities. This design is architecture-agnostic, requires no additional adapter modules, and updates only a small fraction of parameters. Across downstream adaptation and MLLM-DCL experiments, AWARe consistently improves the stability-plasticity trade-off, and remains effective even when using a general-purpose calibration set such as MMMU.

6 Limitations

AWARE relies on a calibration set to estimate activation-based neuron importance. Although our experiments show that a general-purpose dataset such as MMMU can provide a practical substitute when upstream task data is unavailable, the quality and coverage of the calibration data may still affect the selected frozen regions. In addition, our evaluation focuses on representative MLLM backbones and multimodal tasks; broader validation on larger models, more diverse domains, and longer continual learning sequences would further clarify the generality of the approach. Finally, because AWARE explicitly freezes salient parameters to preserve prior capabilities, extremely large distribution shifts may still require tuning the retention ratio to balance stability and plasticity.

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A MLLM-DCL Evaluation Details

We follow the MLLM-DCL benchmark (Zhao et al., 2025) and report four standard metrics for continuous instruction tuning. Let T be the total number of sequential tasks, and let $A_{t,i}$ denote the test accuracy on task i after the model has finished training on task t .

- **Mean Fine-tune Accuracy (MFT).** MFT measures plasticity, i.e., how well the model adapts to each task immediately after learning it:

$$\text{MFT} = \frac{1}{T} \sum_{i=1}^T A_{i,i}. \quad (9)$$

A higher MFT indicates stronger immediate adaptation to the downstream distribution of each new task.

- **Mean Final Accuracy (MFN).** MFN measures the model’s average performance on all tasks after the entire training sequence:

$$\text{MFN} = \frac{1}{T} \sum_{i=1}^T A_{T,i}. \quad (10)$$

Unlike MFT, MFN accounts for later performance degradation, so a high MFN requires both learning new tasks and retaining previous ones.

- **Mean Average Accuracy (MAA).** MAA summarizes the accuracy trajectory throughout continual learning by averaging the cumulative accuracy over observed tasks at each training step:

$$\text{MAA} = \frac{1}{T} \sum_{t=1}^T \left(\frac{1}{t} \sum_{i=1}^t A_{t,i} \right). \quad (11)$$

MAA rewards methods that maintain consistently high accuracy across all known tasks during the learning process.

- **Backward Transfer (BWT).** BWT quantifies the average final change in performance on previously learned tasks relative to their accuracy immediately after learning:

$$\text{BWT} = \frac{1}{T-1} \sum_{i=1}^{T-1} (A_{T,i} - A_{i,i}). \quad (12)$$

A less negative or positive BWT indicates less forgetting, while a negative BWT reflects degradation on earlier tasks.

Method	MFT ↑	MFN ↑	MAA ↑	BWT ↑
ModalPrompt	53.87	53.94	49.67	+0.09
HiDe	60.77	55.31	60.68	-6.82
LoRA-FT	64.98	53.00	61.13	-14.97
MoELoRA	64.94	54.78	61.76	-12.70
CL-MoE	<u>66.06</u>	54.88	61.79	-13.96
O-LoRA	65.16	55.54	62.12	-12.03
SEFE	65.01	58.51	63.63	-8.13
DISCO	64.78	<u>60.33</u>	<u>63.93</u>	<u>-5.57</u>
AWARe	67.49	62.10	67.28	-6.74

Table 7: Detailed CL metric summary on MLLM-DCL.

Training Stage	RS	Med	AD	Sci	Fin
RS	78.87	—	—	—	—
Med	80.76	59.44	—	—	—
AD	80.80	59.03	54.25	—	—
Sci	80.02	56.20	52.94	53.45	—
Fin	78.96	47.01	46.94	46.13	91.45

Table 8: Detailed AWARe accuracy results on MLLM-DCL.

For the detailed AWARe run in Table 8, we use 200 MMMU samples as the calibration dataset when training the first RS task. For each subsequent task, the calibration dataset consists of MMMU together with 100 samples from each previous task, which are used to recompute the activation distributions. Rows indicate the latest task after which the model is fine-tuned, and columns indicate evaluation tasks.

B Additional Model Qwen2.5-VL

To examine whether AWARe remains effective on a stronger recent backbone, we additionally evaluate Qwen2.5-VL (Bai et al., 2025b) on two downstream tasks from MLLM-DCL: Med and AD. We use HallusionBench (Guan et al., 2024), MicroVQA (Burgess et al., 2025), DocVQA (Mathew et al., 2021), VStarBench (Wu and Xie, 2024), and AI2D (Kembhavi et al., 2016) as upstream evaluation tasks to measure retained general multimodal capabilities after task-specific tuning. The zero-shot row reports the original Qwen2.5-VL performance before downstream adaptation.

Across both downstream tasks, SFT reaches the highest target-task accuracy but causes severe degradation on upstream capabilities, especially on DocVQA, VStarBench, and AI2D. In contrast, AWARe consistently preserves the strongest upstream performance among fine-tuned methods while retaining competitive downstream adaptation, indicating that activation-weighted parameter freezing transfers to Qwen2.5-VL and provides a

Method	DCL-Med $\uparrow$	HallusionBench $\uparrow$	Micro-VQA $\uparrow$	DocVQA $\uparrow$	VStarBench $\uparrow$	AI2D $\uparrow$
Zero-shot	0.357	0.770	0.480	0.947	0.620	0.740
SFT	0.521	0.330	0.070	0.599	0.010	0.040
AWARE	0.477	0.760	0.480	0.913	0.650	0.730
SPIDER	0.452	0.670	0.440	0.875	0.580	0.710
LoRASculpt	0.465	0.720	0.460	0.863	0.560	0.690

Table 9: **Qwen2.5-VL results on the MLLM-DCL Med task.** The downstream column reports DCL-Med adaptation performance, while the remaining columns report upstream task retention. Boldface marks the best downstream result and the best upstream retention among fine-tuned methods.

Method	DCL-AD $\uparrow$	HallusionBench $\uparrow$	Micro-VQA $\uparrow$	DocVQA $\uparrow$	VStarBench $\uparrow$	AI2D $\uparrow$
Zero-shot	0.213	0.770	0.480	0.947	0.620	0.740
SFT	0.5595	0.500	0.390	0.190	0.240	0.370
AWARE	0.4629	0.760	0.440	0.901	0.600	0.710
SPIDER	0.432	0.650	0.390	0.856	0.540	0.650
LoRASculpt	0.445	0.690	0.420	0.847	0.520	0.690

Table 10: **Qwen2.5-VL results on the MLLM-DCL AD task.** The downstream column reports DCL-AD adaptation performance, while the remaining columns report upstream task retention. Boldface marks the best downstream result and the best upstream retention among fine-tuned methods.

robust stability-plasticity trade-off.

C Parameter Efficiency Analysis

In this section, we provide a detailed calculation of the trainable parameter ratio in our optimal setting. Consider a standard Transformer block in LLaMA architecture with hidden dimension h . The parameters can be categorized into Self-Attention and Feed-Forward Network (FFN) modules:

- **Self-Attention:** Consists of four projection matrices: $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v, \mathbf{W}_o \in \mathbb{R}^{h \times h}$. The total parameter count is $4h^2$.
- **FFN (SwiGLU):** Comprises three matrices: $\mathbf{W}_{up}, \mathbf{W}_{gate} \in \mathbb{R}^{h_{inter} \times h}$ and $\mathbf{W}_{down} \in \mathbb{R}^{h \times h_{inter}}$. In LLaMA, the intermediate dimension is typically set to $h_{inter} \approx \frac{8}{3}h$. The total parameter count is approximately $3 \times \frac{8}{3}h \times h = 8h^2$.

Thus, the total number of parameters per block is approximately $12h^2$.

In our best experimental setting, we apply the AWARE strategy specifically to the query, key, and value projection layers ($\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v$) with a retention ratio of $\rho = 30\%$, while freezing the remaining components. The number of trainable parameters is calculated as:

$$N_{train} = 3 \times (1 - \rho) \times h^2 = 3 \times 0.7 \times h^2 = 2.1h^2 \quad (13)$$

Consequently, the ratio of trainable parameters to the total parameter count is:

$$\text{Ratio} = \frac{2.1h^2}{12h^2} \approx 17.5\% \quad (14)$$

This analysis demonstrates that our method effectively updates only a quarter of the model parameters, significantly reducing computational overhead while preserving generalization capabilities.

D Experimental Configuration

We summarize the hyperparameter configuration that achieves the optimal performance reported in our main results in Table 11, specifically under the **Global-Highest 30%** setting. All experimental results are averaged over three random runs.

E Baseline Details

Single-task downstream adaptation. We calculate baselines using the following configurations:

- **Full-FT:** Full fine-tuning of the model parameters.
- **LoRA**^[ICLR'22](Hu et al., 2022): Low-Rank Adaptation, which injects trainable rank decomposition matrices.
- **DoRA**^[ICML'24](Liu et al., 2024): Enhances LoRA’s learning capacity and training stability by decomposing weights into magnitude and direction components.
- **DARE**^[ICML'24](Yu et al., 2024): Parameters from the fine-tuned model are randomly selected and rescaled to preserve both generalization and specialization capabilities.
- **Orth-Reg**^[ECCV'24](Hu et al., 2024): Encourages fine-tuned features to remain orthogonal to pretrained features, thereby preserving model generalization.
- **Model Tailor**^[ICML'24](Zhu et al., 2024): Pre-trained parameters are preserved while a small fraction (e.g., 10%) of fine-tuned parameters is replaced based on salience and sensitivity analysis.
- **LoRASculpt**^[CVPR'25](Liang et al., 2025): The method prunes redundant LoRA parameters via sparse updates guided by weight importance and conflict-aware regularization.
- **SPIDER**^[ICML'25](Huang et al., 2025): Updates parameters only where specialization knowledge is more critical than generalization knowledge.

Continual instruction tuning. For the MLLM-DCL experiments, we compare against continual learning baselines that are designed for sequential task adaptation:

Hyperparameter	Value
<i>Knowledge Profiling Phase</i>	
Calibration Dataset Composition	OKVQA, OCRVQA, GQA, TextVQA (200 samples each)
When Upstream Dataset is restricted	MMMU (20 samples each category 600 samples in total)
Selection Strategy	Global Highest Activation
Training Target Components	q_proj, k_proj, v_proj, mm_projector
Retention Quota (ρ)	30% (Top 30% salient neurons frozen)
<i>Downstream Fine-tuning Phase</i>	
Epochs	3
Global Batch Size	16
Learning Rate (LLM Components)	2×10^{-5}
Learning Rate (MM Projector)	2×10^{-4}
LR Scheduler	Cosine Annealing (Warmup ratio: 0.03)
Optimizer	AdamW

Table 11: Detailed Hyperparameter Settings for the Optimal AWARe Configuration.

- **LoRA-FT**^[ICLR’22](Hu et al., 2022): Sequential LoRA fine-tuning on each new subtask without an explicit forgetting-mitigation mechanism.
- **O-LoRA**^[EMNLP’23](Wang et al., 2023): Learns new tasks in orthogonal subspaces to reduce interference with previously acquired knowledge.
- **MoELoRA**^[NIPS’24](Chen et al., 2024a): Uses a mixture-of-LoRA-experts design for continual instruction tuning.
- **ModalPrompt**^[EMNLP’25](Zeng et al., 2025): Introduces dual-modality guided prompts to preserve multimodal knowledge during continual learning.
- **CL-MoE**^[CVPR’25](Huai et al., 2025): Uses a dual-momentum mixture-of-experts mechanism for continual multimodal visual question answering.
- **HiDe-LLaVA**^[ACL’25](Guo et al., 2025a): Decouples multimodal continual instruction tuning hierarchically to mitigate forgetting.
- **SEFE**^[ICML’25](Chen et al., 2025): Separates superficial and essential forgetting to preserve important multimodal capabilities.
- **DISCO**^[ICCV’25](Guo et al., 2025b): Applies federated continual instruction tuning for sequential instruction adaptation.